pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Spread over the Curve

WEEK 3 · DAY 2

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Where does 147 come from?

5.606

Bezane Finance's yield, percent

147

its oas column, basis points

Three steps for every bond

1. **Years to maturity, on 30/360**
2. **The curve yield at those years**
3. **The difference, times 100: basis points**

A spread here describes the data. It is never a view on a bond.

Week 3: curve, spread, and the book

Day 11	Treasury curve interpolation
Day 12	**Spread over the curve**
Day 13	Spread duration and DTS
Day 14	Portfolio and scenarios
Day 15	The capstone

What the spread measures

```
spread_bp = (yield_pct - curve yield at maturity) x 100
```

Bezane Finance, matures 2032-06-18	Percent
Yield	5.606
minus oas, 147 bp	1.47
Curve yield the file used	4.136

One percent is 100 basis points, so the gap in percent is multiplied by 100.

The invented curve, around it

Tenor, years	Yield, percent
3	3.916
5	4.086
7	4.214
10	4.348

4.136 sits between the 5-year and 7-year points.

Years to maturity on 30/360, the file's rule

Count	Days	Years
Actual, over 365	2088	5.720548
30/360, over 360	2058	5.716667

Excel side by side (tested in Excel)

`=YEARFRAC(D2,E2,0)`	5.716667
`=YEARFRAC(D2,E2,0)*360`	2058
`=YEARFRAC(DATE(2027,2,28),DATE(2031,8,31),0)*360`	1621
`=DAYS360(DATE(2027,2,28),DATE(2031,8,31))`	1620

The file read the curve at days_30_360 / 360: years_to_maturity.

YEARFRAC equals years_to_maturity on all 200 bonds. DAYS360 turns that 31st into a 30th: a lookalike, not the file's count.

Bezane Finance, 30 September 2026 to 18 June 2032. Close, not equal: only one reproduces the file.

The curve yield at each bond's maturity

```
# 5.7167 years: between the 5 and 7 year points
share = (years - 5) / (7 - 5)
curve = 4.086 + share * (4.214 - 4.086)
bondmath.interpolate_yield(years, tenors, yields)
```

4.1319

the straight line, percent

4.136

what the file used, percent

The line gives 0.41 bp less. Section 12.6 explains why.

Excel side by side (tested in Excel)

`=MATCH(G2,$A$2:$A$11,1)`	6: the 5-year point
`INDEX(...):INDEX(...)`	the two points
`=FORECAST.LINEAR(G2, yields, tenors)`	4.131867

The full formula is in the notebook. It has no flat ends: #N/A below the first tenor, #REF! at or above the last.

spread_to_curve_bp

```
def spread_to_curve_bp(yield_pct, years,
                       curve_tenors, curve_yields_pct):
    curve_yield_pct = interpolate_yield(years,
        curve_tenors, curve_yields_pct)
    # percent, times 100 basis points per percent
    return (yield_pct - curve_yield_pct) * 100
```

Excel side by side (tested in Excel)

`=(F2-H2)*100`	147.4133

F2 is the yield in percent, as in the file. The *100 turns percent into basis points, as in the function.

147.41

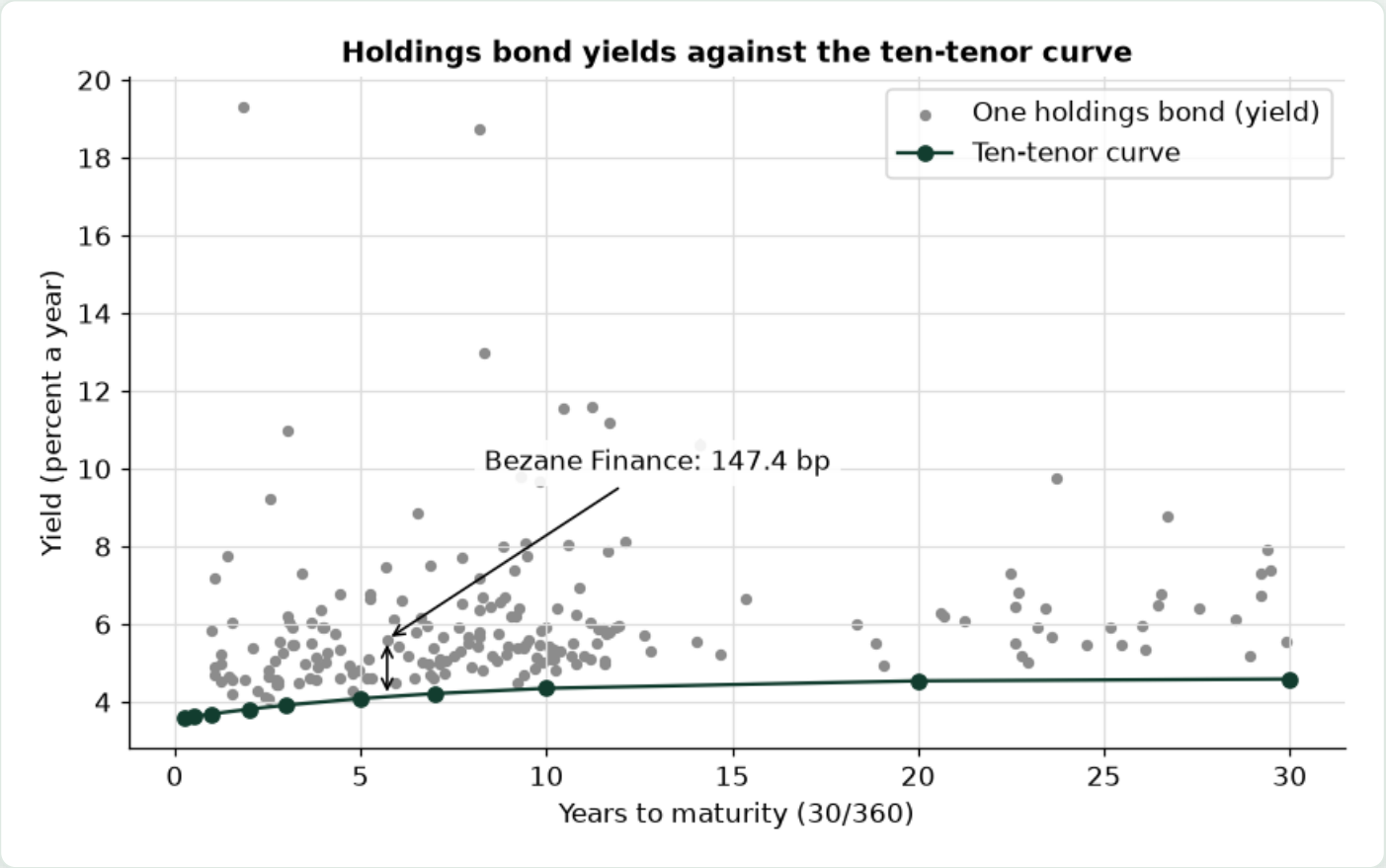
Bezane Finance to the ten-tenor table, bp

147

the oas column, bp

The name says the unit: _bp.

Yields, the curve, and the spread



Each dot is a bond's yield. The line is the curve. The vertical distance from the line up to a dot is that bond's spread.

147.4 bp

Bezane Finance, to the table

The arrow is its spread: the number the function returns.

The oas column, reproduced exactly

```
# the rule the file used (DATA.md)
yield_pct = round(3.55 + 1.05 x (1 - exp(-T / 7)), 3)
            + oas / 100
```

```
# a curve of one point: the bond's own years
spread_to_curve_bp(y, years, [years], [curve])
```

Excel side by side (tested in Excel)

`=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)`	4.136
`=(F2-J2)*100`	147.0000

Column K equals the oas column on all 200 bonds.

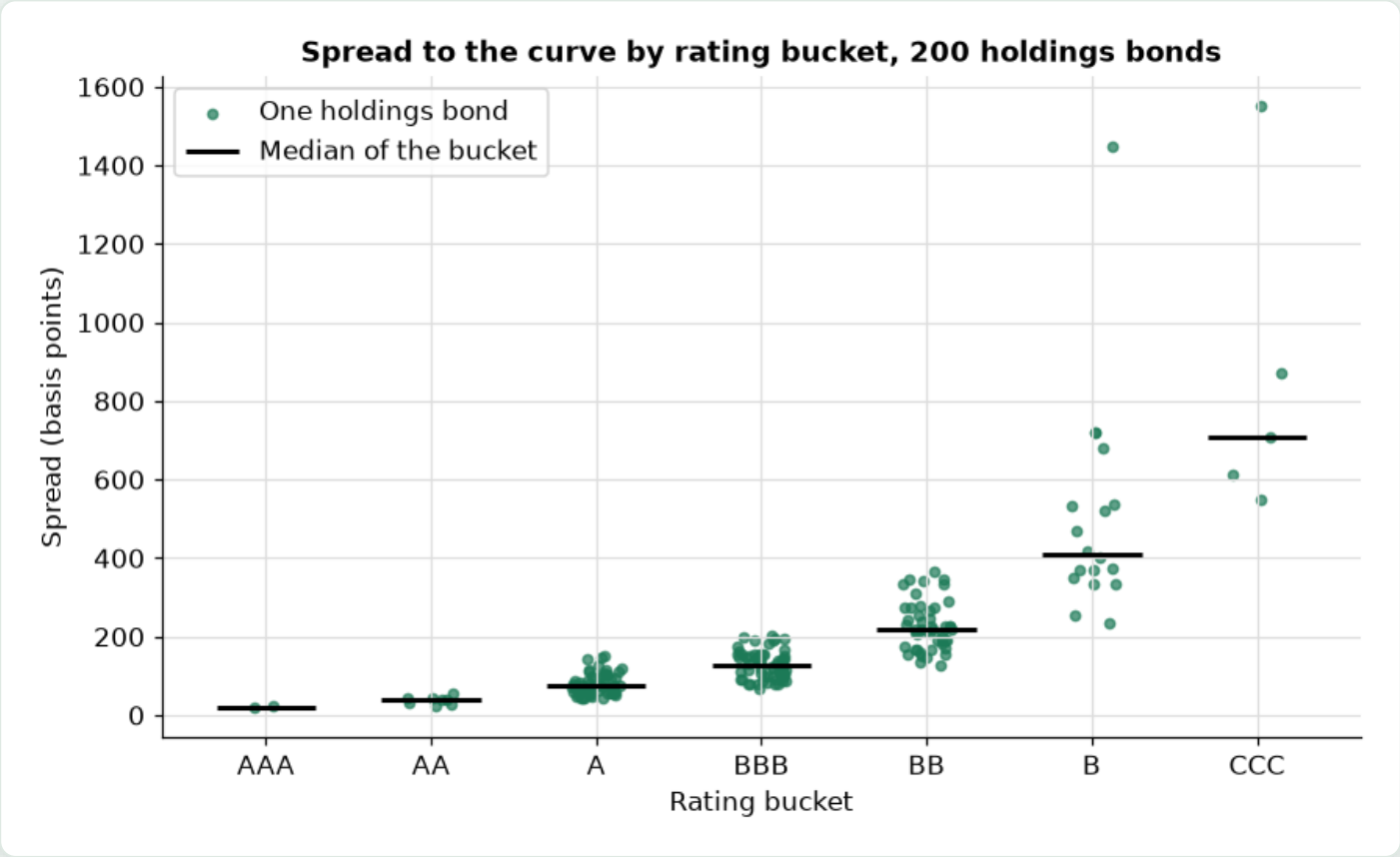
4.136

the formula at 5.7167 years, rounded

2.3e-13 bp

largest miss from oas, 200 bonds

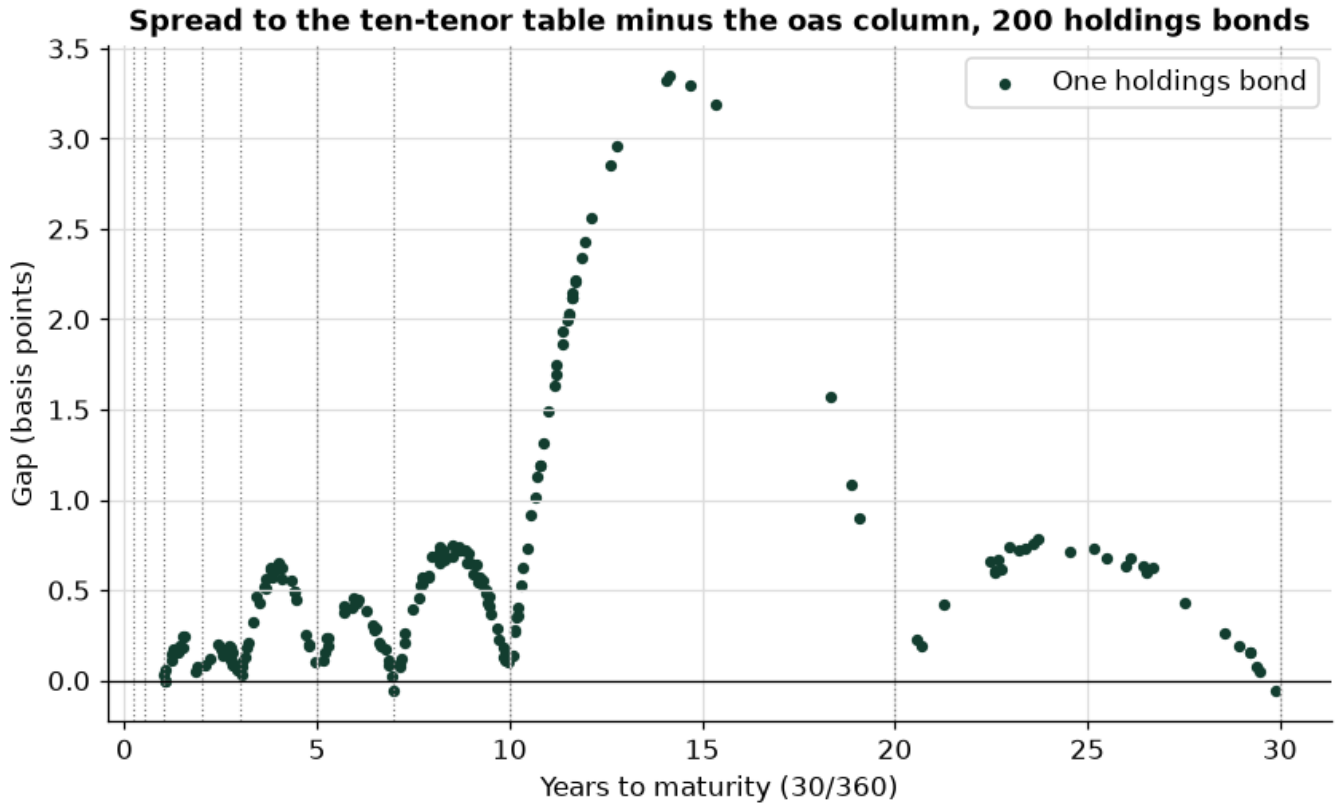
Spreads by rating bucket



Each dot is a bond. The black bar is the median of its bucket.

Spreads rise as the rating falls because the generator that wrote the file was built that way. The picture describes the invented book.

The interpolation choice moves the spread



3.35 bp

largest gap

0.64 bp

average gap

196 of 200: the table's spread is larger

The curve bends. A straight line between two points of it sits below it, so the curve yield is lower and the spread larger.

Compare two spreads only when they were built the same way.

What to call the number

oas

The file's column name. Here, built as a yield spread at the bond's own maturity.

`spread_to_curve_bp`

This course: one yield minus another, times 100. A yield spread.

OAS and Z-spread

On a desk: from a spot curve, each cash flow at its own rate. Beyond this course.

For a bullet with no options, an OAS comes out close to the Z-spread. The course's function is never called oas.

Today's Excel: one sheet, every formula tested

Col	Job	Excel, row 2 (tested)	Bezane Finance
G	Years	`=YEARFRAC(D2,E2,0)`	5.716667
H	Curve, straight line	`FORECAST.LINEAR, INDEX/MATCH`	4.131867
I	Spread to table, bp	`=(F2-H2)*100`	147.4133
J	Curve, the formula	`=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)`	4.136
K	Spread to formula, bp	`=(F2-J2)*100`	147.0000
L	30/360 days	`=YEARFRAC(D2,E2,0)*360`	2058
M	The lookalike	`=DAYS360(D2,E2)`	2058

Curve in A1:B11; settlement in D, maturity in E, yield in percent in F. Fill row 2 down.

How the tests grew

git log --oneline -- test_bondmath.py

```
d58b1a6 Add years_to_maturity and spread_to_curve_...
31ad84c Day 11: interpolate_yield on a curve
cda82e9 End of week 2: dates, accrued, price, dura...
4ae58db End of week 1: price and yield on a coupon...
```

With -- and a file name, git log lists only the commits that changed that file.

git diff --stat week2

```
bondmath.py      | 57 +++
test_bondmath.py | 66 +++
2 files changed, 123 insertions(+)
```

With a tag, git diff --stat counts the lines changed since that commit: days 11 and 12.

The work folder is new each day, so the notebook rebuilds the week1 and week2 tags. Your own folder has them.

Bring it home: paste, test, commit, look back

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Two functions and three tests at the end of your files, without the %%writefile line.

3

Test

python -m pytest -q: 35 passed, or more with your exercise functions.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Look back

git log --oneline -- test_bondmath.py, then git diff --stat week2.

`unknown revision?  git log --oneline, then  git tag week2 <day 10 hash>`

In Colab: download bondmath.py and test_bondmath.py from the work folder. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert against Excel's reference rows. Run it: it fails.

→

4

Fix

Find the convention it broke, fix it, run the test again.

Today: the spread comes back in percent, and the code calls it basis points: 100 times too small.

Code that runs is not the same as code that is right.

What you can do now

- Say what a spread to the curve measures, in basis points
- Count years on 30/360 with years_to_maturity, or YEARFRAC basis 0
- Read the curve with interpolate_yield, or FORECAST.LINEAR and INDEX/MATCH
- Reproduce the oas column exactly; measure the interpolation gap
- Name it a yield spread; see growth with git log -- file and diff --stat

Week 3, Day 3 Spread duration and DTS: how far a price moves when the spread moves